TradeSmart AI

Business Analysis 3.2 · Vaal University of Technology · 2026

16 453 LISTINGS · 2 393 MODELS · 2018–2023

Clarity on every trade-in you value

TradeSmart AI gives South African used-car dealerships an instant, data-driven market value for every trade-in, without the guesswork.

An AI Solution for Industries

XGBoost · NLP · Deep learning

LIVE VALUATION

2019 VW Polo Vivo 1.4 Trendline · 85 000 km

R164 156

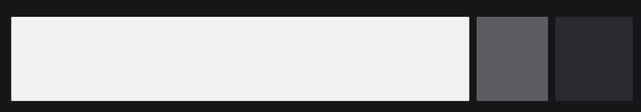
EXACT MODEL MATCH

Range R122 251 – R206 061

- ESTIMATE IN 11 MS
- SAVED TO SQL SERVER
- LOGGED TO EXCEL

MODEL ACCURACY

91.1%	87.8%	R42k
R^2 explained	within ±15%	avg error



75% within ±10% · 13% within ±15% · 12% further

AVERAGE OF 5 RANDOM 80/20 TEST SPLITS

NLP CONDITION DIGEST

NEEDS REVIEW

Respray mentioned after "no accidents"

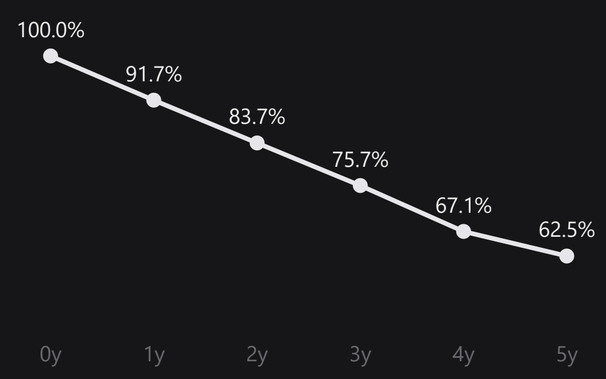
- Dent: small, about the size of a golf ball (87%)
- Aftermarket mods: wheels; reputable installer (82%)
- Respray flagged at 40% confidence
- "Windscreen is fine" ignored

96% CLASSIFIER ACCURACY · 8 CATEGORIES

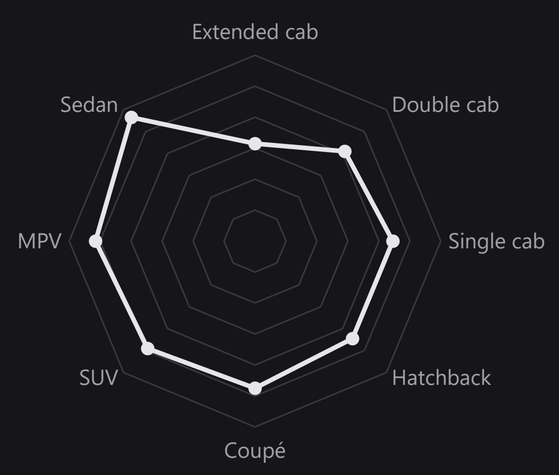
VALUE OVER TIME

−9.5% / year

Average car, like-for-like across model years



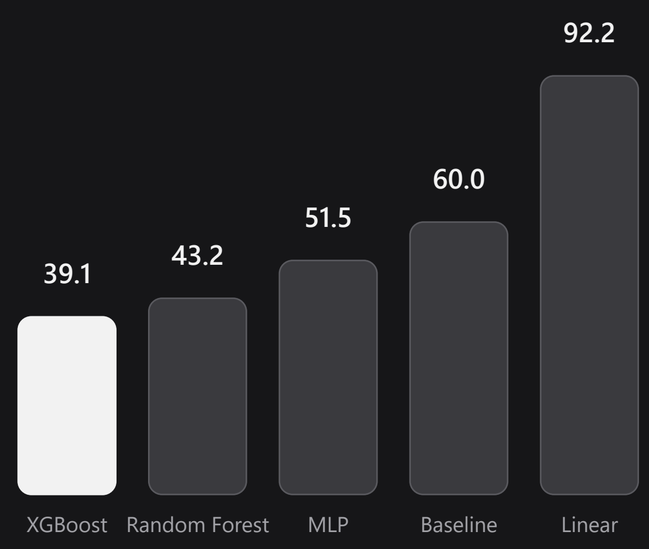
DEPRECIATION BY BODY TYPE



Bakkies hold value best (6.3% a year), sedans worst (11.3%)

MODELS COMPARED

AVG ERROR, R K



THE PROBLEM

Dealerships value trade-ins by hand: slow, inconsistent between consultants, and never recorded. Overpaying loses margin; underpaying loses the deal.

HOW IT WORKS

01 **Chat**
Customer picks their car, year and mileage

02 **Understand**
NLP turns their words into condition findings

03 **Predict**
XGBoost estimates the value and a range

04 **Record**
Saved to SQL Server and an Excel log

Every car, every customer, one clear value

BUILT WITH

Python XGBoost scikit-learn TensorFlow Flask SQL Server GitHub

Group: LC Sekhukhune · H Madzivi · E Harmse · B Malope

Data: Autotrader South Africa listings, published by Adewoyin (2023), MIT licence